Kelvin Schouten

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Profile

MSc Quantitative Finance student with a BSc in Mathematics and six months of independent industrial modelling research. Experience in numerical modelling, Monte Carlo uncertainty analysis, and validation against experimental measurements.

Education

MSc Quantitative Finance 2026–Present
Tilburg University, Netherlands
Relevant coursework: Empirical Finance, Panel Data Econometrics, Valuation & Risk

BSc Mathematics 2022–2026
Utrecht University, Netherlands
BSc research (2025): (Non-)periodicity in fractional order differential systems | Supervisor: Paul Zegeling

HBO Applied Mathematics (not completed) 2019–2021
Higher professional education; continued with a BSc Mathematics

Applied Research Experience

Research and Development: Two-Phase Flow Modelling & Uncertainty Quantification Six-month project
Independent project for an industrial company | Food-industry pipelines

- Reviewed recent scientific literature, implemented and adapted published two-phase-flow models for the company's pipeline equipment and operating conditions.
- Modelled system requirements and cleaning performance across different flow regimes and industrial operating scenarios.
- Implemented Monte Carlo simulations to quantify how uncertainty in model inputs affects predictions of system requirements and cleaning outcomes.
- Validated model predictions against experimental measurements to assess predictive performance under industrial operating conditions.
- Calibrated the model to improve its accuracy and reliability in predicting the emptying and cleaning of the pipelines.

Quantitative Finance Project

Empirical Finance Data Analysis MSc coursework
Python | Fama–French factors and industry portfolio returns

- Processed and analysed Fama-French factor data and industry portfolio returns in Python for regression-based empirical finance exercises.

Technical Skills

Programming & tools: Python, Jupyter notebooks, Git, GitHub

Quantitative methods: Numerical modelling, Monte Carlo simulation, parameter estimation, uncertainty propagation, model validation

Statistics & finance: Regression analysis, econometrics, probability theory, stochastic processes, financial return analysis

Research: Scientific literature review, mathematical model adaptation, technical reporting

Additional Experience

Research Assistant 2019-2019
The Hague University of Applied Sciences

- Analysed student performance data to investigate differences in academic outcomes; documented and presented findings.
- Credited contributor to the 2019 research report *Ondersteuningsnetwerken* for the Inclusive Education research group.

Mathematics & Physics Tutor 2017–2025
Altena College and independent tutoring

- Developed tailored exercises and provided mathematics and physics instruction to secondary school students.

Languages

Dutch: Native | **English:** Proficient